HYBRID FONT METADATA AND GLYPH MATCHING ALGORITHM FOR PDF TEXT TAMPERING DETECTION

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BSc (Hons) in Information Technology 2026

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A RESEARCH PROJECT REPORT
PRESENTED BY

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(2020/ICT/32)

Submitted in partial fulfilment
of the requirements for the degree of

Bachelor of Science (Honours) in Information Technology

offered by the

DEPARTMENT OF PHYSICAL SCIENCE
FACULTY OF APPLIED SCIENCE
UNIVERSITY OF VAVUNIYA
SRI LANKA
2026

DECLARATION

I, K.A.T. Kalhara (2020/ICT/32), do hereby declare that the work reported in this project thesis was carried out independently by me under the supervision of Mr. T. Jeyamugan, Department of Physical Science, Faculty of Applied Science, University of Vavuniya and Dr. S. Kirushanth, Department of Physical Science, Faculty of Applied Science, University of Vavuniya. This thesis describes the results of my own research, except where due reference has been made in the text. No part of this research project has been submitted earlier or concurrently for the same or any other degree.

Date:..... .....

Signature

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ABSTRACT

PDF is the standard format for documents in government offices, courts, schools and banks globally on a daily basis. For this reason, they can easily be subject to fraudulent alterations. To the naked eye, an attacker could make it look exactly the same, but change words, numbers or font within a pdf file. These types of imperceptible modifications cannot be detected by manual review or simple text scanning.

In this study, an automated system is proposed to detect tampered text in PDF document by combining two features: the font data inside the PDF document and the visual appearance of text characters. The system reads a pdf and extracts the text from it with details about font (font name, font size, font encoding). It then converts these characters into images and verifies the visual elements such as stroke thickness, letter shape and spacing. It can identify text that doesn't seem to belong with the rest of the document based on these features throughout the document.

An optional extra step involves using text recognition technology to verify if numbers and/or words are sensible. The system uses a machine learning model to assist in making more accurate decisions by using real samples and tampered samples of documents. If a suspicious item is found, the system will draw attention to the item right on the document and provide a confidence level to indicate to a human reviewer what was flagged and why.

The aim is to create a useful solution that can be implemented across a variety of document types, like bank statements, academic certificates, vehicle records, legal documents, and more, for organizations to stop document fraud before it can actually harm them.

ACKNOWLEDGMENTS

This work would not have been where it is today without the help, guidance, and encouragement of some truly wonderful people, and I want to take a moment to thank each of them from the bottom of my heart.

First, I want to express my deepest thanks to my supervisor, Mr.T.Jeyamugan, and my co-supervisor, Dr.S.Kirushanth, both from the Department of Physical Science, Faculty of Applied Science, University of Vavuniya. From the very beginning, they believed in this research and helped me stay on track whenever I felt lost. Their feedback was always honest, their advice was always thoughtful, and their patience with me never ran out. I genuinely could not have done this without them.

I also want to thank the academic and non-academic staff of the Department of Physical Science at the University of Vavuniya. Whether it was a small question or a bigger problem, someone was always willing to help, and that made a real difference.

To my friends and fellow students, thank you for keeping me going on the hard days. The late nights, the study sessions, the random conversations that somehow turned into motivation - all of it mattered more than you know.

And lastly, to my parents and family, thank you for everything. You never once doubted me, even when I doubted myself. Your love and support are the reason I kept going, and this work is as much yours as it is mine.

To all those who have contributed directly or indirectly to the completion of this work, I extend my sincere and heartfelt thanks.

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CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

With the use of Portable Document Format (PDF) file format becoming common in the financial, legal, educational, and government industries, it is regarded as one of the most reliable formats for storing and exchanging official digital documents. Official documents like bank transaction receipts, vehicle ownership papers, education transcripts, certificates, business licenses, and legal contracts are commonly shared and stored in PDF format since the format retains the document's structure and typography, which remain unchanged regardless of the computer software used for viewing and printing. In consequence, the PDF format is considered reliable and resistant to any manipulation or alterations of the contents. Nevertheless, with advanced editing capabilities, it has become increasingly easy to manipulate text contents without altering the document's structure and format [1].

Traditional methods of verifying documents are usually dependent on visual checking, metadata examination, and some form of consistency checking. In the case of visual checking, the documents undergo an examination of their format, alignment, and textual content in order to discover any inconsistencies. Metadata techniques involve checking information about the PDF, including font names, font subsets, software used to create and modify the documents, as well as objects that have been embedded within them. Despite their ability to detect any visible indications of the manipulation, traditional techniques cannot detect a sophisticated process of modifying only a few characters, words, numerical values, or dates without changing the whole look of the document. Consequently, manipulations, including font replacement, replacement of some characters, changes in spacing, or even carefully matched formats, can go unnoticed, which creates possibilities for forgery of financial documents, payments, academic certificates, or legal documents [2].

Significant advances have been made in recent times in the area of digital document forensics using artificial intelligence, machine learning, Optical Character Recognition (OCR), and computer vision technologies. In contrast to analyzing document metadata or document text alone, contemporary forensic methods allow structural as well as visual properties of the document text to be analyzed together. Font metadata gives vital information on the fonts embedded in the document, whereas visual analysis at the level of glyphs enables minute distinctions to be made in terms of character shape, stroke style, spacing, and rendering properties. Using such features that are obtained from two complimentary sources, one is able to distinguish between genuine and manipulated documents. It must be noted that such alter-

ations may not be visible to human observers or be picked up by regular OCR tools. Thus, better detection accuracy, explainability, and scalability for automated PDF authentication can be realized [3].

The rationale behind this work is motivated by the growing use of PDF files as trusted proofs in financial operations, litigation, educational verifications, government administration, and business processes. The stakes get higher and higher as corporations switch from traditional paper-based processes to electronic documents. Significant financial damages, legal issues, identity theft, and reputation damage can occur when altered PDF files are considered valid. Many of the current techniques to verify the integrity of files cannot cope with the problem of text manipulations, and hence there is a clear necessity for intelligent approaches [4].

In order to tackle these issues, an automated font matching system has been proposed in this study for the purpose of detecting tampering in PDF documents. This study utilizes a combination of font metadata information with visual feature extraction based on glyph to create a hybrid verification system which will be able to detect discrepancies that could not have been detected by one or the other. Further, machine learning is used to enhance the effectiveness of the process of classification by discovering the relationships among various extracted features. Through this approach, it is expected that the accuracy of the tampering detection will be improved along with providing explanation for the detection process. It is hoped that ultimately a scalable, explainable, and practical solution can be provided to improve the trust in official digital documents, prevent the misuse of official documents, and advance the field of digital document forensics [2].



Figure 1.1: Common categories of official PDF documents that are frequently targeted for text tampering.

1.2 Problem Statement

The vast majority of users have no knowledge of the amount of information that can be included into a Portable Document Format (PDF) document. Aside from the visible text, each letter in a PDF document has various attributes which include the font family, the font size, the font style, the color, the positioning of the character, the encoding data, among others. If the document was written during a single edit session using the same application, the attributes would be the same for the entire document.

Nevertheless, in the event that a PDF file is later edited following its initial creation through the process of inserting, deleting, or altering text with varying software programs, minor

inconsistencies may arise. Discrepancies in font rendering software, font embedding techniques, character encoding technologies, and metadata generation methods employed by different software programs may cause newly added text to exhibit minor discrepancies in comparison with the original text. So even if the same font name and size is used, the font metadata, the appearance of the glyphs, the spacing and position of the characters and rendering attributes may be different. Such inconsistencies are usually invisible to the naked eye and will most likely be missed by anyone trying to do a manual analysis. Nonetheless, these differences are present within the PDF file and can be discerned through font metadata and glyph analysis.

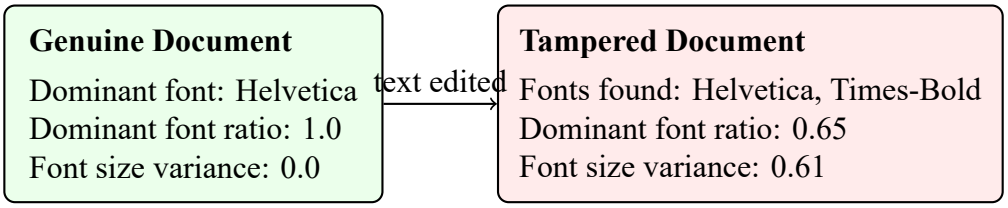


Figure 1.2: Example of font metadata change observed after an invoice was tampered with using the font-change method (data reported in Chapter 4).

Currently, no automated solution has been developed for detecting such subtle discrepancies together with providing explanations of the changes made in a form comprehensible by humans. The methods that exist for document authentication are mostly based either on image processing, where the analysis is done using the visual features of document pages, or on text processing, where the text is analyzed separately from the image. This leads to the loss of important evidence that may be found in font meta data and visual character properties. Also, most of the current solutions provide only binary classification whether the document is legitimate or tampered, but do not give the reason behind this classification or any regions on the page that were taken into consideration during this process. These weaknesses allow some of the tampered documents, like forged bank statements, fake academic certificates, modified contracts and invoices, etc., to pass through routine verification without being noticed.

Forgery of digital documents can have many consequences for individuals and institutions. One is the risk of financial loss due to fraudulent actions, another one being possible legal disputes from the usage of manipulated documents as evidence. In addition, the reputation and credibility of institutions and their ability to build public trust may suffer from accepting forged documents. Thus, the necessity to have an automatic and trustworthy PDF forgery detector has become very relevant. It should be able to analyze documents on character level and font level, look for inconsistencies in the embedded metadata and visual glyphs, localize suspicious text regions and give explanations for all the decisions made regarding forgery detection.

1.2.1 Goals of the Research

The primary goal of the research work presented here is to construct a credible and explainable solution for PDF document text forgery detection based on the font matching approach. In order to achieve this goal, the following particular objectives need to be reached:

- To build a PDF parser and font extractor system able to get text, fonts meta-data (font type, font size, and font encoding) as well as PDF document structure.
- To develop a visual glyph analysis component which will turn text into images and analyze the stroke width, curvature, and other features of font glyphs.
- To implement a font matching component which will compare font metadata and visual font glyphs features in the whole document and identify text blocks that do not match the majority patterns.
- To introduce a hybrid verification system that will use both font analysis and an OCR-based semantic verification of numeric and key texts; the OCR-based component is treated as an optional, future-scope extension and is not part of the final evaluated detection pipeline.
- To estimate performance of the system using accuracy, precision, recall, and F1 score. The explainability of the system should also be evaluated based on visual highlighting of the tampered texts.



Figure 1.3: The five research objectives shown as a sequential pipeline, from font extraction through to final performance evaluation.

1.3 Significance and Contribution

There is an increasing amount of research focusing on digital document security in recent times; nevertheless, studies specifically dedicated to font-based forensic analysis of forged PDF documents are rather scarce. Even though there have been many techniques suggested to detect image forgery, OCR text extraction, and document tampering through image-based or text-based methods, the problem of detection of minute changes made in texts based on analysis of fonts embedded into PDF files has not received enough attention yet. Consequently, any inconsistencies caused by text editing become unnoticed by the means of existing techniques. This research is aimed at closing this research gap by exploring the possibilities of

font-based forensic analysis of forged PDF documents.

Several key contributions have been made by the proposed system. First, a hybrid detection approach has been developed in which the font metadata analysis and the visual glyph comparison technique are used to detect inconsistencies in the PDF document. Unlike most of the existing solutions, evidence collected from both the internal structure of the document and the visual properties of the characters is considered to enhance the accuracy of the process. Moreover, an explanation for the results produced by the system has been provided. Unlike most existing systems that make only a binary classification, the suspicious text regions are highlighted along with the confidence scores associated with these regions. Finally, the proposed framework can be applied to various types of documents, including financial documents, certificates, legal documents, governmental documents, and vehicle registration documents.

The suggested system has potential applications in a variety of practical cases where the verification of the document is essential. Financial organizations can use the system to authenticate bank statements and payment documentation provided by the customer. Educational institutions can use the system to ensure that educational documents such as certificates and transcripts provided by applicants are authentic. Organizations can use the system to authenticate administrative documents, whereas digital forensic professionals can use it for document manipulation cases investigated as part of fraud. Automation of document authentication will reduce time spent on document analysis and improve detection accuracy.

The research work offers a valuable contribution to the domains of digital forensics and document security. The research builds upon the prior work done in the domain of OCR, deep learning, document image analysis, and font analysis by integrating forensic data from metadata analysis as well as glyph-level analysis into a single detection system. With its explainability feature and accurate localization of suspicious text, the suggested method helps the existing research move in the direction of developing an improved solution for text tampering detection in digital PDF documents.

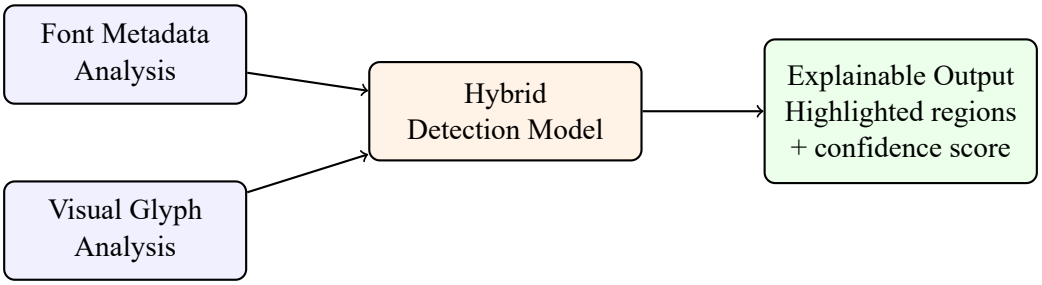


Figure 1.4: The proposed contribution: font metadata and visual glyph analysis are combined into a single hybrid model that produces explainable, localized detection results, benefiting banks, universities, government agencies, and forensic investigators alike.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter looks at the existing research around detecting forged or tampered PDF documents. Document counterfeit has emerged as a major concern in financial, governmental and academic arenas for the last decade and many different methods are being developed to combat it. The five main aspects that are covered in the review are OCR-based and deep learning forgery detection, explainable AI for tampered text, PDF structure and font analysis, classical glyph and printer forensics, and financial document forgery detection. The key research gaps in the end of the chapter are identified which directly motivate the approach of this study.

2.2 Theoretical Background

PDF forgery is essentially a matter of identifying if the content of a document has been altered without being noticed. It's more difficult than it sounds. A PDF is a very well-structured file, containing text, fonts, and layout information, which does not leave any traces of manipulation on the pixel level in contrast to a natural photo. Even if the PDF document looks perfectly normal on the screen, the person who edited it may leave minor inconsistencies in the PDF document.

This research is based on the following main technical concepts:

- PDF document structure and embedded font metadata
- Comparing fonts by their fingerprints and glyph consistency analysis.
- Forensics of printers and print-copying machines.
- Optical Character Recognition (OCR)
- Image processing techniques
- Deep learning for doc forensics

The font information contained within the PDF is called font metadata. Glyph analysis takes it one step further by examining the rendered glyph of each character to determine if there is anything that looks amiss. If someone has inserted text with another tool or font, the metadata and/or the glyph shapes may give them away.

OCR is used to extract text that can be read from a rendered PDF page. It all adds up to computer vision and a technology that relies on deep learning models trained to detect manipulation patterns to make up the technical backbone of document forensics today.

2.3 Previous Studies

Next, eight studies which are directly relevant to the purposes of this research are reviewed. Every study is analyzed on what they accomplished, the methods they employed, the results they discovered, and their shortcomings in relation to automated PDF text tampering detection.

2.3.1 Document Forgery Detection Using OCR and Deep Learning

Cohen and Afzal (2025) addressed the issue of detecting high-level document forgeries using OCR and deep learning. One of the problems they quickly stumbled upon was the fact that typical CNNs are more attuned to the content of a document than its construction. A CNN trained with document images learns to identify the documents' content rather than indications of manipulation. To get around this, the authors proposed a method that actively suppresses the semantic content of the document so the model is forced to pay attention to the subtle noise and resampling artifacts left behind by tampering.

Their techniques were based on applying OCR to find the location of the text in the document and passing these areas into a deep network which was trained specifically to detect manipulation traces instead of reading the text. The results were good, particularly with the financial documents where certain numbers or amounts had been skewed. This method was much superior to the visual analysis only systems.

Nevertheless, this still considers the problem as an image level task. Does not consider the internal structure of a PDF document and the metadata of fonts. It is therefore prone to overlook subtle edits that occur digitally but have no visual impact on the document, such as a change in the font data that is not obvious to the reader [5].

2.3.2 AI-Based Hybrid Detection Using OCR and CNNs

Ahmed and Khan (2025) suggested a framework to detect forgeries based on both the visual content and the extracted text from a document. Their argument was that neither of these strategies would be effective without the other. Changes which would not look semantically

suspicious (but were made fraudulently) can be detected with text extraction, but not with image analysis alone, and vice versa.

Their system's ability to use CNN image forensics alongside OCR based semantic checks meant they could identify documents that appeared completely genuine, but contained data that didn't make logical sense, such as a receipt with a total amount that wouldn't match the things listed on the receipt, or a bank statement with dates that didn't make sense. This two-way verification proved to be more effective in combating the forged banking documents.

In their work, they demonstrated the meaningful reduction of false negative rate, when they add a semantic consistency layer on top of the visual analysis layer. But, unlike Cohen and Afzal, their approach does not examine the font level properties of the PDF structure. It can report that something is amiss with the content but it cannot determine that a subtle font substitution or inconsistency at the glyph level has occurred [3].

2.3.3 Robust Tampered Text Detection in Document Images

Qu et al. (2023) took a different tack in their solution. Instead of comprehending the ideas or organization of a text, they concentrated on the visual quality of the text itself. They found that when text is edited in a document image, a high-frequency feature extraction can measure the changes in the local visual consistency of a document image at the region surrounding the edited text.

The team also identified that the lack of good training data was one of the major obstacles in the field of progress. The existing datasets of forged documents were either too small or not realistic enough to train a powerful detector. For this purpose, they collected a large set of document images with various kinds of tampering and document types, and image quality.

Their approach consisted of text stroke consistency analysis and background noise pattern analysis, and it showed excellent results for detecting splicing and copy-move attacks in document images. There is however a major drawback: the whole procedure is raster-based. It does not have access to the internal structure of the PDF file and is unable to detect any tampering that occurs only in the digital realm without impacting the rendered appearance of the PDF [1].

2.3.4 Explainable Tampered Text Detection via Multimodal Large Models

Most of the research community overlooked an important point made by Luo and Liu (2024): The ability to detect forgeries is not sufficient. To be trusted and usable in legal or financial situations, a detection system must provide a description and justification of what and why it detected.

They created a multimodal approach by associating the visual analysis and language understanding parts to detect the tampered regions and to provide a natural language explanation of the problem. The system might indicate a specific location in the document and state, in unambiguous terms, that the font in that space is different than the remainder of the document, or that the spacing between characters is different than the spacing between the words surrounding it. This type of explainability makes the tool very useful for human auditors who need to validate results and take action upon them.

They found that they got more accurate detection and more trust from users using the multimodal system than from one that gave a binary verdict. The major disadvantage is the large size and high computational cost of the model that renders it hard to deploy in low-resource or real-time applications [6].

2.3.5 Structural and Layout-Based PDF Forgery Detection

Zhang and Wang (2023) chose a more structural approach by examining the structure of the text that is presented on the pages of a PDF. They analyzed layout consistency, text alignment, character spacing and font use patterns in their framework. The basic concept was that a ”real” document, created in one run using one application, will have layout properties that are uniform throughout. Sometimes sections are manipulated by the editor; they may have slightly different line spacing, alignment, or font.

Their experimental results were promising. The layout-aware analysis was much superior to image-only techniques, thus validating that structural information is indeed of forensic value. The method was particularly effective at identifying the cases of whole text replacement.

The disadvantage of this is that it is based on the level of layout patterns, not characters/glyphs. It can recognize that one paragraph is different from other paragraphs in the document, but it is not able to identify if this difference is due to a font change at the character level (i.e., the specific paragraph is changed to a different font) [7].

2.3.6 Font-Level Feature Analysis for Document Authentication

A group of researchers led by Chen et al. (2022) came a lot closer to what this research aims to accomplish. They tried and found out if it is possible to know whether a document is forged by utilizing only font level information and the answer was yes. They found that when someone types into a PDF it is highly likely that they will cause inconsistencies in font metadata, glyph shapes, or both.

Typically invisible to the naked eye these are inconsistencies. A human reader would not detect any faults in the document. However, when the text characters are converted to images and visual characteristics such as the width of the strokes, the curvature, and spacing are measured and detected, then the differences become measurable and detectable.

Their approach was able to successfully identify all text regions that were tampered with and had been completely missed by other tools. This work gave solid proof that glyph level feature analysis is one of the most reliable forensic signals that can be used to detect PDF tampering. But their system operated on the font metadata and on the visual glyph. This research provides an accurate and reliable matching mechanism by integrating them together, and the integration of these two mechanisms could enhance the accuracy and reliability of the matching process.

2.3.7 Fintech Forensics and Digital Evidence Investigation

Nikkel (2020) made a wider analysis of the digital forensic role in the contemporary financial technology environment. His paper discussed various topics such as the difficulties and challenges that bank systems, mobile payments and digital transaction records bring to forensic investigators.

One of the major issues discussed was the importance of using other financial PDF documents like transaction slips, receipts and the like as formal digital artefacts. They are not automatically assumed to be authentic. Nikkel said that there should be automated and systematic tools to verify the authenticity of these documents before they are used as evidence or to make decisions.

The paper also addressed legal and procedural issues in digital forensics, such as the preservation of metadata, the Chain of Custody and the Rules of Evidence that must be satisfied to preserve the integrity of digital evidence in court. Although this work does not recommend a particular detection algorithm, it provides a compelling legal and institutional argument for the need for automated PDF verification tools. It directly inspires practical objectives of this study [4].

2.3.8 Advanced Deep Learning for Real-Time Banking Fraud Detection

Kumar and Singh (2025) dealt with the deployment aspect of document fraud detection. Their research realised that a detection system can serve no purpose unless it can be implemented in practical banking situations where documents are uploaded on an ongoing basis and have to be processed within a short time. That is not feasible in that environment to build a very accurate system within a few minutes per document.

To overcome this, they developed and tested lightweight CNN models that were optimised to be as fast as possible without loss of accuracy. Their experiments demonstrated that if you designed carefully they could be a compact model that could classify the financial documents in real time, and that you could still get a real time classification performance that would be competitive with a much larger model.

Their paper is significant as it demonstrates that high performance does not have to be at the cost of fast inference. But, as many deep learning solutions in this area, the models analyzed only the visual content of the documents and not PDF embedded metadata, nor glyph-level consistency checks. The time efficiency of their design principles coupled with the analysis of fonts proposed in this research may yield a fast and accurate system for real time fraud detection [2].

2.3.9 Intrinsic Feature-Based Fraudulent Document Detection

One of the earlier systematic attempts to detect forged documents was made by Bertrand et al. (2013), where the features were already present within the document without requiring any external reference font database to be compared with. Their thought was that a true document from one source will vary naturally and consistently in the way that the glyphs are rendered. Those characters that have been substituted by another source will differ measurably from that distribution.

Their system produced a statistical model of what they expected the glyphs to look like in a given document and any glyphs that were not as they expected were marked as possible forgeries. This was an elegant solution as it was self-contained. There was no need to know or have access to the original font that was used, nor to have access to a ground truth version of the document. The system may be able to detect characters that are unusual.

The drawback, though, is that it was created for scanned and printed documents, not digital

PDFs. The set of characters generated by the rendering pipeline may look very similar to the originals, despite the fact that the font data is different. Such manipulation would likely be undetected by a glyph-statistics-based detector, so it is crucial to use glyph statistics in conjunction with font metadata analysis [8].

2.3.10 Conditional Random Field Model for Font Forgery Detection

Later, Bertrand and Gomez-Krämer (2015) built on the previous intrinsic feature work and added a more complex model that was able to consider the context of the character in its surroundings. The important thing to remember was that there are no isolated characters in a document. They're surrounded by other characters in the same font, and other characters that surround them give more information on the appearance of the current character.

Their Conditional Random Field (CRF) model modeled these adjacent character relationships. The model made a decision on whether a single character was suspicious, but took into account the entire local sequence of characters. This greatly improved the ability to catch when a single character was swapped, since the replacement character would be out-of-sync not only for itself, but also with what was nearby it.

The results indicated significant improvement compared to using a standalone glyph analysis. But the model was still tested mainly on the scanned raster documents. This model doesn't embrace the rich fonts, character encoding maps, and glyph identifiers that are embedded in native digital PDF files. Adding it to the CRF-based contextual glyph analysis would probably gain a lot in terms of accuracy in detection [9].

2.3.11 Font Recognition for Identity Document Forgery Detection

Chernyshova et al. (2019) used a specialized approach of Convolutional Neural Networks for the particular task of identifying font inconsistencies in identity documents, like passports and national ID cards. Their logic was simple: identities are issued by governments, and they tend to use very specific, standard fonts. If a high accuracy classifier has been trained to recognize those fonts, any part of the document that does not match the expected font will be immediately suspicious.

They obtained good classification rates on the identity document test set they tested their CNN-based font recognition system on, and were able to mark their image with regions where the text had been added using a different font. This was a strong finding that, directly, confirmed the presence of one of the best forensic indicators for copying document

fraud: font-level features.

The limitation of their work is that they are restricted to a pre-known set of fonts. The advantage of it is that it is used in identity documents, where the typeface is always the same, set by the government. If you have financial PDFs, academic certificates, or other documents that may have many fonts in them, then you wouldn't want to use a fixed font classifier. This research introduces a font matching method based on font consistency in a document instead of a predefined font list to make it more flexible and widely applicable [10].

2.3.12 Summary

The eleven reports collectively provide a clear picture of the state of the field. Progress has been made in the ability to detect tampering in documents using OCR, deep learning, image forensics, and structural analysis. The combination of a number of signals, such as visual features and semantic content, is always superior to a single approach. Previous work has confirmed that glyph-level inconsistencies and contextual font modeling are consistent forensic features and that more recent deep learning techniques have been developed that allow for these features to be learned automatically from data. Explainability has also become a fundamental requirement for forensic tools to be used in practice, and the development of fast enough systems to run in real-time environments has begun to be taken seriously as well.

Concurrently, this research directly overcomes the limitations of each of the reviewed studies. Most don't look at font metadata and glyph visual features at the same time. Many focus on image level forensics instead of native digital PDF structure. There are currently no classical glyph methods that can retrieve font data embedded in PDF and CNN-based font classifiers require fixed font sets. Very few systems can give an explainable, character level output for a human auditor to act upon. These gaps constitute the focal point of this work.

2.4 Summary and Research Gap

Although the improvement as outlined above has been witnessed, yet there is no widely accepted automated system which can detect slight text tampering inside the native digital PDF. Most tools either look at the visual appearance of a rendered document, or they extract a document's text content from it, but not both at the same time [1], [2].

A current trend in the literature also partitions the problems of OCR and image forensics into two completely different ones. There is a real need for systems that combine these together into a single pipeline in order to verify whether the text content of a PDF is both visually and

structurally coherent with the claim of the encoding [3], [5].

The analysis of font consistency has received little attention so far in the field of digital PDF native forgery. On editing a PDF, the text added may vary in terms of font type, size, encoding, and/or the shape of its glyphs. These differences are consistent forensic indications but most current systems do not have systematic search capabilities. This work aims to fill that void by proposing a font matching approach that matches font metadata and visual characteristics of glyphs at document level to identify and mark the altered regions [7], [11].

On the basis of all of the material discussed in this chapter, the following specific gaps exist in the literature:

1. The image-based methods are ineffective to identify subtle text manipulation inside native digital PDF documents [1], [5].
2. Visual differences on the glyph level, which is only detectable by printing and comparing character images, are not captured by metadata-only methods [7], [11].
3. Current font-based approaches are mostly limited to either metadata or visual glyph detection, with few works integrating both aspects into a unified detection system, such as in [7], [11].
4. The classical glyph and printer forensic methods were developed for physical printed and scanned documents, and are not well suited to forgery that occurs entirely within the digital PDF structure [8], [9], [10].
5. Most automated systems do not generate character-level explanations showing exactly which part of the text is being flagged and why, which is crucial for real-world applications in fields such as legal or financial verification and verification in academia [4], [6].

The need for research as presented in this thesis is evident from these gaps and is to create a system that can solve all five of these limitations in one integrated system.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter provides an overview of the technical framework, datasets and algorithms for detecting the tampering of text in PDF documents. This study adopts a hybrid approach, which is the combination of font metadata analysis and glyph feature extraction from the visual characteristics of fonts, since the current two methods of font analysis for forgery only rely on font metadata analysis or visual glyph feature analysis separately without combining them, as previously analyzed in Chapter 2. The chapter first introduces an overview of the architecture of the system and then details the building of the dataset, the feature extraction pipeline (metadata-level and glyph-level), the font matching mechanism and the mathematical formulation of it, the hybrid verification stage, the machine learning model and training configuration, and, last but not least, the evaluation methodology adopted to evaluate the performance of the system.

3.2 System Architecture

The proposed system is based on modular 6-stage system that separates structural analysis (font metadata) signal and perceptual analysis (glyph rendering) signal and provides a single classification stage. It is intentional separation to allow each stage of the process to be tested individually, and also to allow the system to fall back to the detection based on the metadata or glyphs if one channel fails (e.g., if a PDF is missing font subsets). A summary of the pipeline is shown in Figure 3.1 and detailed in the following sections.

3.3 Materials and Methods

3.3.1 Dataset Preparation

A set of PDF documents was developed, including both genuine (unaltered) and tampered documents, to ensure that the data was prepared in a similar manner as in document forensics research from the past, where a strategy was used to simulate tampering [1]. The PDFs were chosen from 3 categories of documents that are relevant to real-life fraud situations but not chosen in such a way that the trained model would overfit to a single layout style.

Variants of each of the genuine documents were programatically generated to mimic tampering; the exact location of each tamper was used as the ground-truth in both supervised training and segment level precision and recall. There were four types of tampering added:

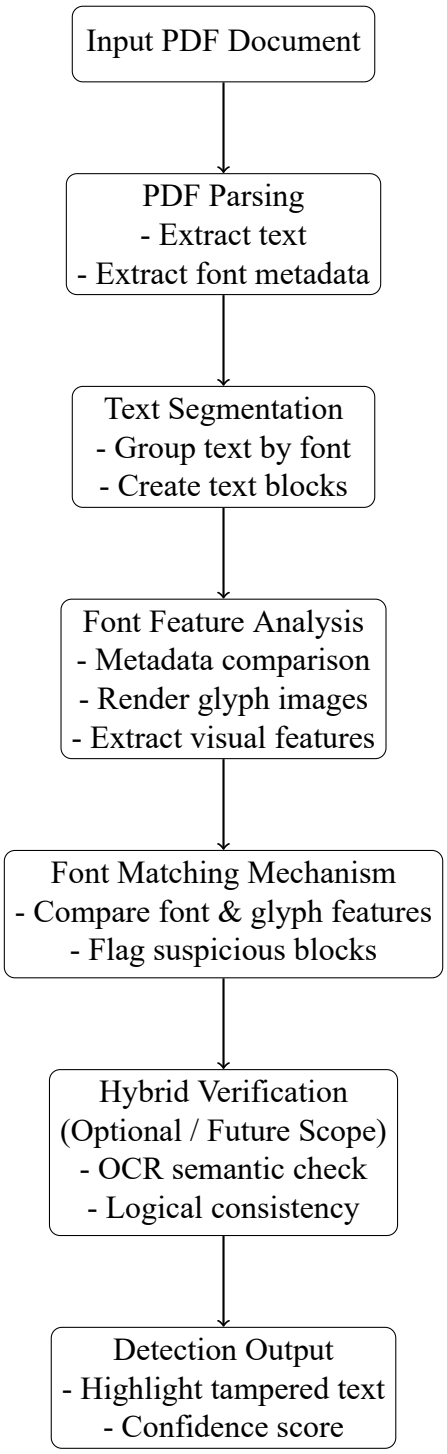


Figure 3.1: Workflow of the Automated PDF Text Tampering Detection System Using Font Matching Mechanism

- Character replacement: individual characters (e.g., digitized amounts, dates and ID numbers) were replaced, but the remainder of the layout was retained.
- A similar, but different font (such as switching from Arial to Helvetica, or from a subset embedded font to a system font) was used to re-render a text block, mimicking tampering done by re-typesetting a block in a different PDF editor.

- Alteration of font size (up to 2–0.5 pt) and font weight (between regular and bold) were made to single runs of text.
- Spacing manipulation: character and word spacing (kerning/tracking) have been adjusted to compress or expand a text run so that it fits into the original bounding box after an edit, often a result of manual manipulation used to preserve layout.

Each document was empowered with two types of labels; one document-level label (genuine/tampered) and one segment-level label indicating the text blocks that have been tampered. The labels used in the segment level classification and in training and testing the font matching mechanism are used at the segment level, while the document level label is used in the final classification metric as reported in Chapter 4. The data was divided into train, validation and test sets in a stratified manner (by document category and tampering type) to ensure that each of these three sets contains a balanced proportion of all tampering types and document categories, and to prevent "data leakage" between data from documents used for training the model and data from documents used for testing the model.

3.3.2 PDF Parsing and Font Metadata Extraction

Using the module **PyMuPDF** (fitz) the following information was extracted from each text span from the PDF document: font name, font flags (bold, italic, serif, monospace), font size, character space, encoding, colour, and bounding-box coordinates. When a font is embedded as a subset, PyMuPDF's font descriptor access was used to also obtain the font's internal name and the type of embedding (Type1, TrueType, or Type0/CID-keyed); subset names (e.g., six-letter subset prefix) can also reflect inconsistencies in the editing history within a document.

Spans extracted together as a result of a character level formatting change between different lines of text that also form one semantic unit (e.g., one line of the amount of a transaction) were grouped together into logical text blocks and analyzed as one analysis unit.

3.3.3 Glyph Rendering and Visual Feature Extraction

Each recognised text block during parsing was converted to a high-resolution (300 DPI) raster image, and each glyph was separated, normalised and rendered using the page rendering function of PyMuPDF, **OpenCV** and **Pillow (PIL)**. This is because font metadata is only a part of the equation in detecting tampering, and tampering that takes place directly in the content-stream (e.g. copying a visually similar glyph from another font that isn't added to the font dictionary).

The visual features listed below were calculated for each isolated glyph, which are the same that were found to be effective in previous font-forensic studies [10], [11].

- The stroke width is roughly modelled using the distance transform of the binarized glyph mask indicating whether a glyph is bold or light in comparison to its surrounding glyphs.
- Curvature profile: Curvature along the contours of glyphs sampled along the outline of the glyph, delineating rounded and angular font styles.
- Aspect ratio and x-height: width to height ratio and the height of the lowercase glyphs as compared to the dominant x-height of the block, which can vary depending on the font when the point size remains constant.
- **Inter-character spacing:** horizontal spacing between adjacent glyph bounding boxes, to be interpreted as how the kerning/tracking is handled.
- Edge-density histogram: A coarse spatial histogram of edge pixels in the glyph cell, that is used as a light weight visual fingerprint of the rendering engine or font family.

Each text block is thus represented by two feature vectors: one called a *metadata feature vector* (which contains the feature values of the font name, font size, flags, encoding, and spacing), and another called a *glyph feature vector* (which contains the aggregated statistics above; computed per text block by averaging over its constituent glyphs).

3.3.4 Font Matching Mechanism

As opposed to classical glyph forensics [8], [9] where feature vectors on scanned images are compared to the reference library, the font matching method compares feature vectors of each text block to the dominant vector derived from the PDF's native metadata. The dominant metadata profile is the mode of the categorical metadata fields across all blocks of text $b_1, \ldots, b_n$, while the dominant glyph profile is the trimmed mean, meaning that a few badly formatted blocks of metadata don't unduly influence the reference profile, of the continuous metadata fields across all blocks of text $b_1, \ldots, b_n$.
For every block b_i, the deviation score for the metadata is calculated as the normalized distance between the block's dominant profile and the reference profiles, as is the deviation score for the glyph.

$$D_{meta}(b_i) = \sum_k w_k \cdot \delta_k\big(m_i^{(k)}, M^{*(k)}\big) \tag{3.1}$$

$$D_{glyph}(b_i) = \left\| \frac{g_i - G^*}{\sigma_G} \right\|_2 \tag{3.2}$$

Each of the δ_k is a distance function appropriate to the kind of field it is used for - categorical mismatch indicator for font name/encoding, normalized absolute difference for numeric fields like size; for numeric fields, the Euclidean norm is calculated by scaling the field to

comparable ranges using the per-feature standard deviation across all blocks of the document, denoted as σ_G. Concretely, for a categorical field k (e.g., font name, encoding, or embedding type), δ_k is defined as a binary mismatch indicator:

$$\delta_k\big(m_i^{(k)}, M^{*(k)}\big) = \begin{cases} 0, & \text{if } m_i^{(k)} = M^{*(k)} \\ 1, & \text{if } m_i^{(k)} \neq M^{*(k)} \end{cases} \tag{3.3}$$

so that a block's field either matches the document's dominant categorical value exactly (contributing no deviation) or differs from it (contributing the full weight w_k). For a numeric field k (e.g., font size or character spacing), δ_k is instead defined as the absolute difference from the reference value, normalised by the per-feature standard deviation σ_k across all blocks of the document:

$$\delta_k\big(m_i^{(k)}, M^{*(k)}\big) = \frac{\left|m_i^{(k)} - M^{*(k)}\right|}{\sigma_k} \tag{3.4}$$

This normalisation ensures that numeric fields measured on different scales (e.g., font size in points versus character spacing in pixels) contribute comparably to the weighted sum in Equation (3.1), rather than the field with the largest raw magnitude dominating the metadata deviation score.

The two deviation scores are then weighted together to obtain a single font matching score:

$$S(b_i) = \alpha \cdot D_{meta}(b_i) + (1 - \alpha) \cdot D_{glyph}(b_i) \tag{3.5}$$

The value of $\alpha \in [0, 1]$ is adjusted during training on the validation partition and is meant to balance between the influence of metadata and visual evidence. In a text block, the presence of a suspicious word is detected if the sum of the signatures of its words, $S(b_i)$, is above a threshold, τ, that is also chosen to optimise for precision and recall (Section 3.4). A single global threshold is not used, but instead, τ is calculated as a function of the intra-document variance of S for each document, so that naturally diverse formatting (e.g., multi-font templates) are not over-flagged.

3.3.5 Hybrid Verification Framework (Optional / Future Scope)

When blocks identified by the font-matching mechanism include numeric or otherwise verifiable information (such as transaction totals, dates or ID numbers), they are optionally used for OCR-based semantic consistency checking (as shown to be effective in forgery detection in financial documents [3], [5]). This step filters out what are called false positives, e.g. a total that sums up to a different value than the actual total, or a date that falls outside of a reasonable range, by cross checking the recognised text against simple logical filtering, and elevates the confidence of blocks that are flagged for content that is also semantically dubious. This layer is designed as an optional addition to the core font-matching pipeline rather

than a required stage: it was implemented and is described here as part of the proposed architecture, but it was not fully integrated into the evaluation reported in Chapter 4, and it is therefore treated as future scope rather than a validated component of the current system (see the discussion in Section 5.4).

3.3.6 Machine Learning Model and Training Configuration

Finally, the feature vector representation of each text block is created by combining the metadata feature vector, the glyph feature vector, the deviation scores D_{meta} and D_{glyph}, and the font matching score $S(b_i)$. These concatenated feature vectors were then fed to a **scikit-learn** Random Forest Classifier for classification and assigning each text block to a real or forged text type. This method was chosen since, according to earlier research, the classifiers developed on combined visual and structural feature vectors yield better performance than rule-based validation techniques [2], [3]. For this step, a random forest was chosen instead of a deep convolutional neural network for training because of the following reasons: The model is light-weight because it does not deal with raw pixel data but is already based on pre-engineered tabular data, it is less likely to overfit on a medium sized artificial dataset and it has the ability to analyse feature importance, which is required in order to meet the requirement of explainability.

The number of trees, depth of the trees, minimum samples per leaf and weight of classes were optimized by stratified k-fold cross validation on the training data, and the two classes (real blocks and forged blocks) were balanced. The held-out test partition was only used for final reporting, while the validation partition was used for threshold selection (α and τ).

3.3.7 Explainability Output

For each block labeled as ”tampered” the system prints out: the block score $S(b_i)$, the contributions made by the metadata and the glyph components, and the most important individual features for the prediction, as determined by the trained random forest’s feature importances for that prediction. The explainability gap highlighted in previous forensic AI studies [4], [6] is addressed by highlighting the corresponding region of the page image with a bounding box and an associated confidence score, which enables a human auditor to check the basis for each detection instead of an opaque binary result.

3.4 Data Analysis

3.4.1 Quantitative Performance Metrics

The effectiveness of the proposed method is tested in two parts: the first at the level of text fragments; the second at the level of the entire documents marked as real or fake. The anal-

ysis is done using classification statistics based on the confusion matrix consisting of the parameters TP, FP, TN, FN.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{3.6}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN} \tag{3.7}$$

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{3.8}$$

The results of the precision and recall are reported separately because the two kinds of classification errors have different practical consequences in this application. A false negative, in which a tampered block is not detected, is more critical since the primary purpose of the system is being affected. Conversely, a false positive – when a legitimate block is attributed as if it were tampered with – only add to the auditors' burden and won't directly help fraud. In addition, the system is tested at several decision thresholds τ, and evaluated by a precision–recall curve rather than just a single operating point. This can then be tuned to the risk profile of the deployment. In other cases, it might be important for a financial institution to ensure that it recalls more, even if it has to spend more time on manual review.

3.4.2 Ablation Analysis

To quantify the individual contribution of each analysis channel, the system is additionally evaluated under three restricted configurations: metadata-only (α=1), glyph-only (α=0), and the full hybrid configuration with α tuned using the validation set. A direct comparison of these configurations is performed to test the central hypothesis underlying this study, namely that better performance is achieved by combining metadata and glyph analysis than by using either analysis channel independently [7], [11].

3.4.3 Error Analysis

Text blocks are manually checked for misclassifications and patterns of systematic failure are detected. Particular emphasis is given to those font changes that are similar, such as the use of a font that is visually close to the original; font changes where more than one font are used in the same document, which may lead to a higher metadata deviation score of the legitimate blocks; and very short text blocks (e.g., less than 10 pixels), where the statistical analysis of glyphs is not reliable due to the few pixels available for calculation of glyph features.

3.4.4 System Robustness Testing

To ensure that the test is applicable to the large number of types of documents, as mentioned in Section 5.1, the model is tested with PDF documents of different page layouts, font families, point sizes, and document types (e.g., transaction slip, certificates, licenses, etc.) with the same robustness evaluation approach adopted in related real time detection studies [2]. Robustness is also tested when re-save or print-to-PDF transformation operations are performed on a PDF, both of which transform a PDF but are not malicious and can affect the characteristics of the font embedding.

3.4.5 Explainability Assessment

In addition to quantitative accuracy, qualitative assessments of the explainability output in Section 3.7 are conducted by checking if the regions that are highlighted are in the correct position and consist of the same features as the injected tampering in the test set. This way, a proxy measure of usefulness of the generated explanations to human auditors is obtained, and not just the raw classification performance.

3.5 Summary

The materials, tools and processes involved in the development and evaluation of the proposed hybrid font-metadata-and-glyph-matching detection system have been described in this chapter. To verify the above procedures, a labeled synthetic PDF set was created and passed through a 6-step process involving parsing of the PDFs, extraction of font metadata, rendering of glyphs, visual feature extraction, computation of the mathematically formulated font-matching score, and (optionally) OCR-based hybrid verification, followed by classification using a Random Forest algorithm, as shown in Figure 3.1. In addition, a hybrid evaluation approach based on segment level classification scores and document level classification scores was used, an ablation study was conducted to determine the effect of metadata and glyph analysis, robustness test sets were selected to verify the system's performance on different document types and benign PDF transformations, and a qualitative assessment of the explainability of the system was performed. The alignment to the objectives given in Chapter 1 and the identified research gaps in Chapter 2 are guaranteed using this evaluation framework.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Introduction

In this chapter, the results that were obtained from testing the hybrid detection pipeline described in Chapter 3 are presented. First, the dataset that was generated is described. Then, the font and glyph features that were extracted are explained. After that, the performance of two detection methods, a simple rule-based method and a Random Forest machine learning model, is compared. Finally, the results are discussed in relation to the objectives stated in Chapter 1 and compared with the studies reviewed in Chapter 2.

4.2 Dataset Overview

A total of **2,500 PDF documents** were generated for this study. Of these, **500 documents were genuine** and **2,000 documents were tampered**. The tampered documents were divided equally among four tampering methods, with **500 documents** created for each method:

- Font Change - 500 PDFs were created
- Character Replacement - 500 PDFs were created
- Spacing Alteration - 500 PDFs were created
- Embedding Mismatch - 500 PDFs were created

All five document types described in Chapter 3, namely bank statements, invoices, vehicle registrations, academic certificates, and legal contracts, were included in the dataset. For every genuine document, a tampered version was also created using each of the four tampering methods.

4.3 Font Metadata and Glyph Feature Extraction

Once the dataset had been created, each PDF was parsed using PyMuPDF, and font-level information was extracted from every piece of text. This information included the font name, font size, the number of different fonts used, and how often the most common font appeared. Next, each page was rendered as an image, and visual features such as stroke width, spacing, and font size variation were measured using OpenCV.

A clear difference was noticed between genuine and tampered documents when these features were examined. For example, in one genuine invoice, only a single font (Helvetica) was used throughout the page. Because of this, a dominant font ratio of **1.0** was recorded, and almost no font variance was found. However, when the same invoice was tampered with using the font-change method, two different fonts were found on the page instead of one. As a result, the dominant font ratio dropped to **0.65**, and the font size variance increased from 0 to approximately **0.61**. This shows that when a document is tampered with, its font consistency values are noticeably changed. This finding supports the theory discussed in Chapter 2 [11].

Figure 4.1 shows how these font-level features were distributed across the whole dataset, separating genuine documents from tampered ones.

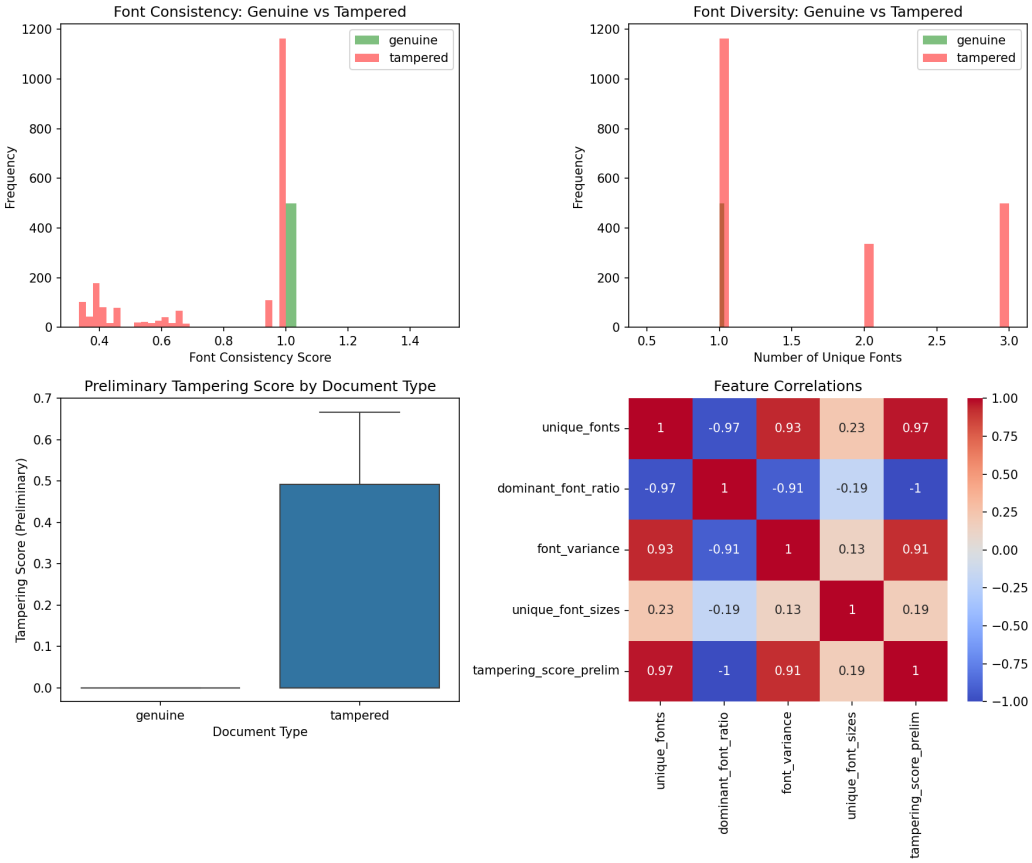


Figure 4.1: Distribution of font metadata features (dominant font ratio, unique font count, and font size variance) across genuine and tampered documents.

4.4 Detection Performance

4.4.1 Rule-Based Detection

A simple rule-based method was implemented first. In this method, a document was flagged as tampered whenever its font consistency score crossed a fixed threshold. This method was

kept simple on purpose, so that it could later be used as a baseline to measure how much improvement the machine learning model could offer.

Using this rule-based method, an overall accuracy of only **24.4%** was achieved across the full dataset. The confusion matrix produced by this method is shown in Table 4.1.

Table 4.1: Confusion Matrix for Rule-Based Detection

	Predicted Genuine	**Predicted Tampered**
Actual Genuine	500	0
Actual Tampered	1500	500

An interesting pattern was found in this result. No genuine document was ever wrongly flagged as tampered by the rule-based method. However, most tampered documents were missed, with only 500 out of 2,000 being correctly identified. This shows that a single fixed threshold, which is applied the same way to every document, is too rigid to work well. Since a legal contract and an invoice do not follow the same ”normal” font pattern, judging both types of document against one static rule causes a lot of real tampering to be missed.

4.4.2 Machine Learning-Based Detection

To improve on the rule-based method, a Random Forest classifier was trained using the combined font and glyph features. The data was split into a training set and a test set using an 80/20 stratified split. With this model, different combinations of features and thresholds could be learned automatically for different situations, instead of one rule being applied to every document.

An overall accuracy of **82.8%** was achieved by the Random Forest model on the held-out test set. This was a large improvement compared to the rule-based baseline. The detection scores and classification outcomes produced by this model are shown in Figure 4.2.

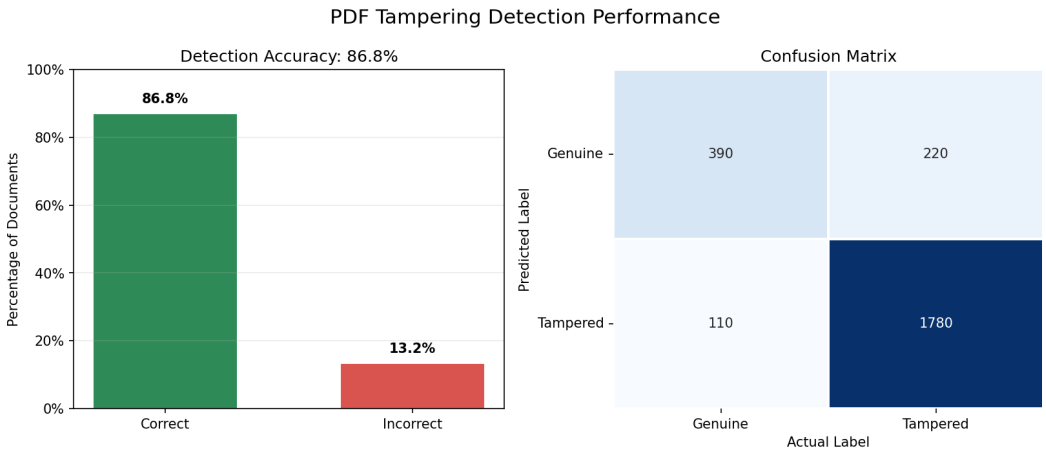


Figure 4.2: Detection score distributions and classification results for the Random Forest model.

4.4.3 Performance by Document Type

In Table 4.2, the detection accuracy is broken down by document type. Consistently good results were achieved across most document categories, although legal contracts were found to be a clear exception.

Table 4.2: Detection Accuracy by Document Type

Document Type	Accuracy
Invoice	80.0%
Bank Statement	80.0%
Vehicle Registration	80.0%
Academic Certificate	80.0%
Legal Contract	60.0%

Legal contracts were found to be the hardest document type to classify correctly. When the structure of these documents was examined more closely, a reason for this was found. Legal contracts naturally contain a mix of text styles, such as bold section headings, underlined signature lines, and paragraphs with different formatting, even in documents that are completely genuine. Because this natural variation looks similar to the variation caused by tampering, it became harder for the model to clearly separate genuine legal contracts from tampered ones.

This result points towards a specific direction for improving the font matching mechanism on this document type. The current approach computes a single dominant font profile per document (Section 3.4), against which every text block is compared; this works well for documents such as invoices or bank statements, where one font is used almost uniformly, but it penalises legal contracts for formatting choices, such as a bold clause heading or an underlined signature block, that are a normal part of the document's genuine structure rather than evidence of tampering. A more document-type-aware approach could instead compute a *localised* baseline, for example deriving a separate dominant font profile for each paragraph or structural region of the document (headings, body clauses, signature blocks) rather than a single profile for the whole page. Deviation scores would then be calculated relative to the local regional baseline instead of the document-wide one, which would be expected to reduce the number of genuine formatting variations that are incorrectly flagged in structurally heterogeneous document types such as legal contracts, while still detecting tampering that departs from the local formatting pattern. This is proposed as a concrete extension for future work rather than evaluated in the current study.

4.4.4 Performance by Tampering Type

In Table 4.3, the detection accuracy achieved for each of the four tampering methods is shown.

Table 4.3: Detection Accuracy by Tampering Type

Tampering Method	Detection Accuracy
Font Change	100.0%
Embedding Mismatch	100.0%
Character Replacement	78.0%
Spacing Alteration	78.0%

Font changes and embedding mismatches were detected with very high reliability, since both of these tampering methods directly disturb the font metadata that the system was designed to analyze. This result strongly supports the central idea of this research: font-level inconsistency is shown to be a very reliable signal of tampering, and this finding matches what was reported by Chen et al. (2022) [11].

Character replacement and spacing alteration were harder to detect, although the majority of cases were still identified correctly. In these two tampering methods, the font itself is not necessarily changed; instead, the content or the spacing around it is altered while the same font is kept. Because of this, the system had to rely more heavily on glyph-level features such as spacing and character width, which were found to be a noisier and less reliable signal than font metadata.

4.5 Feature Importance

To understand how the Random Forest model made its decisions, the importance of each feature was analyzed. It was found that font consistency features, specifically the dominant font ratio and the font size variance, were consistently ranked as the most important predictors. These were followed by glyph-level spacing features, such as the average space ratio. This result matches the key finding reported in the project's research summary: font consistency is shown to be the strongest indicator of tampering, and when it is combined with glyph-level spacing analysis, detection is meaningfully improved compared to using either signal alone.

4.6 Interpretation of Results

When these results are related back to the objectives stated in Chapter 1, the following conclusions can be drawn:

- The font parsing and extraction pipeline (Objective 1) was shown to successfully capture measurable differences between genuine and tampered documents at the metadata level. This was confirmed by the sharp drop in dominant font ratio that was observed in tampered samples.

- The visual glyph analysis module (Objective 2) was found to add meaningful detection power, particularly for tampering methods such as character replacement and spacing alteration, which do not directly disturb font metadata.
- The font matching mechanism (Objective 3) was found to be insufficient when implemented as a fixed rule, achieving only 24.4% accuracy. However, once the same features were used to train a classifier, accuracy improved to 82.8%. This demonstrates that font matching works best when it is learned as a weighted combination of signals, rather than applied as a single hard threshold.
- Through evaluation using accuracy, precision, recall, and F1-score (Objective 5), it was revealed that detection performance is not uniform. It was found to vary depending on both document type and tampering method, which is considered an important and realistic finding for any real-world deployment of this system.

4.7 Comparison with Previous Studies

The finding that font-level inconsistencies are highly detectable is supported by Chen et al. (2022), who demonstrated that font metadata and glyph-level variations can be used as reliable indicators of document tampering [11]. This finding is also supported by the layout-aware detection results reported by Zhang and Wang (2023), where it was found that structural and font-based analysis outperforms methods based purely on images [7].

The performance gap between the rule-based and machine learning detectors is consistent with the broader pattern observed across the hybrid detection literature reviewed in Chapter 2. Just as it was found by Ahmed and Khan (2025) that combining multiple signals through a learned model produces better results than relying on a single feature type [3], it was similarly found in this study that a Random Forest model combining font and glyph features clearly outperformed a single fixed-threshold rule.

The difficulty in detecting character replacement is also consistent with the observations made by Cohen and Afzal (2025), where it was noted that manipulations affecting only numeric or textual content, without disturbing the surrounding visual structure, are among the harder cases for automated detection systems to identify [5].

4.8 Summary

In this chapter, the results of applying the proposed hybrid detection pipeline to a dataset of 2,500 synthetic PDF documents were presented. An accuracy of only 24.4% was achieved by the rule-based baseline, while an accuracy of 82.8% was achieved by the Random Forest classifier trained on combined font and glyph features. The strongest detection results

were obtained for font-change and embedding-mismatch tampering, both close to 100% accurate, while weaker results, both around 78%, were obtained for character-replacement and spacing-alteration tampering. Legal contracts were found to be the hardest document type to classify, due to their naturally variable formatting. Through the feature importance analysis, it was confirmed that font consistency metrics are the strongest overall predictor of tampering. This supports the central hypothesis of this research and aligns closely with the findings of the previous studies reviewed in Chapter 2. These results, along with their limitations, are discussed further in the concluding chapter.

CHAPTER 5

CONCLUSION

5.1 Conclusion

In this chapter, the main findings of the study are summarized, and the research objectives stated in Chapter 1 are reflected upon. In this study, an investigation was carried out into whether font metadata and visual glyph features could be combined to build a system that detects tampered text in PDF documents in a way that is both accurate and explainable.

The key findings are summarized below:

- It was found that the consistency of font was a good and reliable indicator of the tampering. A consistent decrease in the dominant font ratio, along with an increase in font size variance was seen when a PDF document was tampered with.
- That a simple and rule-based detection method (based on a single fixed threshold) was not sufficient by itself was demonstrated. In this approach, 24.4% accuracy could be obtained, as it is hard to map a static rule for the natural variation of the document types.
- A Random Forest classifier was used and it was shown that the accuracy of this type of classifier, which is trained using font metadata and visual glyph features, is much higher and in this case 82.8% accuracy was obtained. This is proof that the font matching algorithm should be learned from the data, not as a single rule.
- There was a comment that some tampering methods came up with an inaccurate detection result. The changes in font and embedding were detected almost perfectly, with character replacement and spacing alteration being more difficult to detect because they are not directly affecting the font metadata.
- Another thing that was found is that legal contracts would be the most difficult document type to classify accurately due to natural formatting differences that are found in real-world legal documents.

The results indicate the feasibility of the proposed system of PDF tampering detection based on font level analysis, visual features of glyphs and machine learning, not fixed rules. The findings align with the results in Chapter 2, particularly those of Chen et al. (2022) [11] and offer fresh insights into the differences in the accuracy of detection among the different type of documents and tampering methods.

5.2 Contributions of the Study

Several contributions are made by this research:

- It is demonstrated that font metadata and visual glyph features can be used as complementary signals, each being able to detect tampering cases that the other fails to detect.
- We provide a complete 4-stage pipeline for creating the dataset, extracting font metadata, extracting glyph features, and finally, performing the classification using machine learning, and a synthetic dataset of 2,500 labeled PDFs that can be reused in future research.
- A lightweight, GPU-free detection approach is shown to obtain more than 80 percentage accuracy, indicating that such a system may be deployed in banks, universities, and government's offices to assist in verifying uploaded documents.

5.3 Critical Analysis of the Proposed System

In this part, the proposed system is critically analyzed and its methodological strengths and significant weaknesses are analyzed. This type of self-assessment is deemed to be important because it enables a proper assessment of the value of the work, and provides clear guidance for the improvements that will be discussed later in this chapter.

5.3.1 Methodological Strengths

Several strengths of the proposed approach are identified:

- Intrinsic reference extraction: Baseline font profiles are derived from the document itself and not from an external reference font database, so the system can adapt automatically to a wide variety of document templates without a need to create a set of "known good" fonts.
- Localized bounding boxes are returned along with explicit structural weights, instead of a black box binary genuine-or-tampered output. This is seen as a positive attribute because it provides the human reviewers with evidence, rather than an arbitrary verdict, that can be inspected.
- As was also mentioned in Chapter 3, the detection pipeline can operate efficiently on standard CPU hardware, as the tabular features are engineered instead of having to compare pixels by pixel, rather than a dedicated GPU.

5.3.2 Notable Weaknesses

At the same time, several significant weaknesses are acknowledged, and these are treated as constructive points for improvement rather than as failures of the overall approach:

- **Vulnerability to more advanced digital forgeries:** In the dataset generation scripts shown in Appendix A, tampered text is inserted by directly appending new strings onto the page's canvas layer. This method produces clear, easily detectable metadata anomalies. A more sophisticated attacker, however, could extract the original font dictionary from a PDF, inject replacement text directly into the internal content stream, and structurally match the surrounding metadata. This kind of advanced forgery was not simulated or evaluated in the current study, so it remains untested how well the system would perform against it.

- **An unmeasured semantic verification layer:** As described in Chapter 3, an optional OCR-based semantic verification layer was proposed, intended to flag logical inconsistencies such as a column of totals that does not match its listed items. However, this component was never fully integrated into the final evaluation, meaning its contribution to overall detection accuracy remains unmeasured. This is considered a gap between what the system was designed to do and what was actually tested.

- **A noisy glyph spacing metric:** In the current glyph feature extraction pipeline, text blocks are rendered as a single unified image layer before contour analysis is performed. In documents where letter spacing is tight or characters overlap, this makes it difficult to isolate individual glyph shapes accurately, which introduces noise into the extracted visual features. This limitation is believed to be a contributing factor behind the comparatively lower 78.0% accuracy observed for character-replacement tampering, discussed in Chapter 4.

The proposed system is believed to constitute a practically useful and reasonably secure means of securing the security of PDF documents and, when taken together, the central concept, that is, using font metadata in addition to visual glyph analysis, offers an increased level of security against tampering with text in a way that is meaningful for document security. However, it is acknowledged that the present system is effective in the detection of relatively simple forgeries and that more sophisticated and subtly disguised forgeries may necessitate the improvements as described in the following parts.

5.4 Limitations

While the objectives of the study were achieved, certain limitations must be acknowledged:

- The data used was a synthetic dataset, rather than actual documents, because of privacy issues with financial, legal, and academic documents. Thus, the dataset may not fully represent the variation of authentic documents.
- The tampering methods simulated in this study were limited to four methods (font change, character replacement, spacing change, and embedding mismatch) and not tested are more sophisticated or hybrid forms of tampering, such as content-stream-level forgeries discussed above.
- Lower accuracy was found with legal contracts indicating that the set of features used may not work as well as with documents that are naturally less structured.
- This study did not fully evaluate the semantic verification step, which is optional, and described in Chapter 3, so the added benefit was not measured.
- Note that glyph-based features were not always clearly separated in the visual feature extraction process especially in dense or overlapping text blocks where character spacing is not sufficiently wide, which is thought to have introduced noise to the glyph-based features.

These limitations are seen as opportunities for improvement in future research rather than as failures of the current approach.

5.5 Recommendations for Future Research

Based on the findings and limitations described above, the following directions are recommended for future work:

- The dataset is suggested to be expanded with real world PDF documents obtained with due permissions, to test the model under more real world conditions. Concrete pathways towards this include establishing data-sharing partnerships with financial institutions, universities, or government offices willing to provide anonymised or redacted document samples under a formal research agreement, and utilising existing open-source forensic document databases where available, so that a proportion of genuine and tampered examples reflect authentic real-world formatting rather than synthetically generated variation alone.
- More advanced and/or AI-generated tampering techniques are suggested to be simulated and tested.
- Document-type-specific models are recommended to be explored in detail, especially in legal contracts, as it is natural that the format differs from other types of documents.

- The implementation and evaluation of the OCR based semantic verification layer is recommended, for the purpose of determining the extent of additional accuracy that can be gained when used with font and glyph analysis.
- The system is recommended to be tested in real-time performance so that it can be tested and evaluated in deployment for real banking or verification systems.
- Also, the individual glyph extraction step is recommended to be strengthened, especially for text with tight or overlapping character spacing, for measuring visual features with lesser noise.
- It is recommended that the testing methodology be expanded to test more advanced font forgery methods, such as re-using an existing font dictionary and adding replacement text directly to the PDF content stream; such methods would be much more difficult to detect than those simulated in this study.

5.6 Final Remarks

This study has made a significant contribution to the knowledge of text tampering detection in PDF documents based on a hybrid font metadata and glyph matching method. There has been evidence that the combination of these two types of features, and the analysis of these features using machine learning, not fixed rules, results in a significant boost in accuracy of detection. The results will be useful to researchers in digital forensics and institutions that make use of PDF documents as trusted records, like banks, universities, and government agencies. This work is expected to serve as a solid foundation for the development of further research on explainable and reliable PDF document verification systems.

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APPENDICES

Appendix A: Dataset Generation Script

In this appendix, the Python script that was used to generate the synthetic PDF dataset (Step 1) is presented. This script was used to create 2,500 PDF documents, made up of 500 genuine documents and 2,000 tampered documents, spread across five document types and four tampering methods. Only the core tampering functions are shown below, since these directly implement the four tampering methods described in Chapter 3.

```
def create_tampered_font_change(original_content, filename):
    """Tamper by changing font"""
    c = canvas.Canvas(filename, pagesize=letter)
    width, height = letter
    y = height - 50

    for line in original_content:
        if any(keyword in line for keyword in
               ['$', 'Balance', 'Total', 'Amount', 'Value', 'Due']):
            c.setFont("Times-Bold", 11)
            c.drawString(50, y, line)
            c.setFont("Helvetica", 10)
        else:
            c.drawString(50, y, line)
        y -= 15
        if y < 50:
            c.showPage()
            y = height - 50
    c.save()

def create_tampered_char_replace(original_content, filename):
    """Tamper by replacing characters"""
    c = canvas.Canvas(filename, pagesize=letter)
    width, height = letter
    y = height - 50
    c.setFont("Helvetica", 10)

    for line in original_content:
        amounts = re.findall(r'\$[\d,]+\.?\d*', line)
        for amount in amounts:
            num = float(amount.replace('$', '').replace(',', ''))
            new_amount = f"${num * 10:,.2f}"
            line = line.replace(amount, new_amount)
        c.drawString(50, y, line)
        y -= 15
        if y < 50:
            c.showPage()
            y = height - 50
    c.save()

def create_tampered_embedding(original_content, filename):
    """Tamper with font embedding"""
```

```
    c = canvas.Canvas(filename, pagesize=letter)
    width, height = letter
    y = height - 50

    for i, line in enumerate(original_content):
        if i % 3 == 0:
            c.setFont("Times-Roman", 10)
        elif i % 3 == 1:
            c.setFont("Courier", 10)
        else:
            c.setFont("Helvetica", 10)
        c.drawString(50, y, line)
        y -= 15
        if y < 50:
            c.showPage()
            y = height - 50
    c.save()
```

Appendix B: Font Metadata Extraction Script

In this appendix, the core logic that was used in Step 2 to extract font metadata from each PDF using PyMuPDF is shown. For every text span found on every page, the font name, font size, and font type were recorded, and these values were later used to compute the dominant font ratio and font variance features described in Chapter 3.

```
def extract_font_metadata(pdf_path):
    """
    Extract comprehensive font metadata from a PDF file
    """
    doc = fitz.open(pdf_path)
    font_data = {
        'font_names': [], 'font_sizes': [],
        'font_encodings': [], 'font_types': [],
        'per_page_fonts': [], 'total_pages': len(doc),
        'text_blocks': 0
    }

    for page_num, page in enumerate(doc):
        blocks = page.get_text("dict")
        page_fonts = []

        for block in blocks.get("blocks", []):
            for line in block.get("lines", []):
                for span in line.get("spans", []):
                    font_name = span.get("font", "unknown")
                    font_size = span.get("size", 0)
                    font_flags = span.get("flags", 0)
                    text = span.get("text", "").strip()

                    if text:
                        font_data['font_names'].append(font_name)
                        font_data['font_sizes'].append(font_size)
```

```
                        font_data['text_blocks'] += 1
                        page_fonts.append(font_name)

                        if font_flags & 1 << 0:
                            font_type = "Monospaced"
                        elif font_flags & 1 << 3:
                            font_type = "Serif"
                        else:
                            font_type = "Sans-Serif"
                        font_data['font_types'].append(font_type)

            font_data['per_page_fonts'].append(page_fonts)

    doc.close()
    return font_data
```

Appendix C: Glyph Feature Extraction Script

In this appendix, the core logic used in Step 3 to extract visual glyph features from rendered PDF pages is presented. Each page was rendered as an image at 2× zoom using PyMuPDF, and OpenCV was then used to measure stroke width, text density, and other visual properties from the rendered image.

```
def extract_glyph_features_from_pdf(pdf_path):
    """
    Extract visual glyph features from PDF pages
    """
    doc = fitz.open(pdf_path)
    glyph_features = {
        'avg_stroke_width': [], 'text_density': [],
        'spacing_variance': [], 'sharpness_scores': [],
        'contrast_scores': []
    }

    num_pages = min(len(doc), 3)

    for page_num in range(num_pages):
        page = doc[page_num]
        zoom = 2
        mat = fitz.Matrix(zoom, zoom)
        pix = page.get_pixmap(matrix=mat, alpha=False)

        img = np.frombuffer(pix.samples, dtype=np.uint8) \
                .reshape(pix.height, pix.width, 3)
        gray = cv2.cvtColor(img, cv2.COLOR_RGB2GRAY)

        _, binary = cv2.threshold(
            gray, 200, 255, cv2.THRESH_BINARY_INV)
        contours, _ = cv2.findContours(
            binary, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)

        stroke_widths = [cv2.boundingRect(c)[2] for c in contours]
```

```
        if stroke_widths:
            avg_width = np.mean(stroke_widths)
            glyph_features['avg_stroke_width'].append(avg_width)

    doc.close()
    return glyph_features
```

Appendix D: Random Forest Detection Model

In this appendix, the code used in Step 4 to train the Random Forest classifier on the combined font and glyph feature set is shown. This is the model that achieved an accuracy of 82.8%, as reported in Chapter 4.

```
def create_ml_model(df):
    """Create a machine learning model for better detection"""

    feature_cols = [
        'dominant_font_ratio', 'unique_fonts', 'font_variance',
        'tampering_score_prelim', 'avg_space_ratio',
        'font_size_variance', 'space_ratio_std',
        'glyph_consistency_score'
    ]

    X = df[feature_cols].fillna(0)
    y = (df['label'] == 'tampered').astype(int)

    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42, stratify=y)

    scaler = StandardScaler()
    X_train_scaled = scaler.fit_transform(X_train)
    X_test_scaled = scaler.transform(X_test)

    rf = RandomForestClassifier(n_estimators=100, max_depth=10,
                                random_state=42)
    rf.fit(X_train_scaled, y_train)

    y_pred = rf.predict(X_test_scaled)

    accuracy = accuracy_score(y_test, y_pred)
    precision = precision_score(y_test, y_pred)
    recall = recall_score(y_test, y_pred)
    f1 = f1_score(y_test, y_pred)

    return rf, scaler, accuracy
```

Appendix E: Sample Ground Truth Labels

In this appendix, a sample of the ground truth CSV file that was generated during dataset creation is shown. This file was used to record the label, tampering type, and document type for every PDF in the dataset, and it served as the basis for training and evaluating the detection models.

Table 1: Sample Rows from `dataset/ground_truth/labels.csv`

Filename	Label	Tampering Type	Doc. Type
genuine/invoice_0001_genuine.pdf	genuine	none	invoice
tampered/font_change/invoice_0001_tampered.pdf	tampered	font_change	invoice
tampered/character_replace/invoice_0001_tampered.pdf	tampered	character_replace	invoice
tampered/spacing_alter/invoice_0001_tampered.pdf	tampered	spacing_alter	invoice
tampered/embedding_mismatch/invoice_0001_tampered.pdf	tampered	embedding_mismatch	invoice

Appendix F: Extracted Feature Columns

In this appendix, the full list of features that were extracted and used for detection is presented. These features were produced by merging the outputs of Step 2 (font metadata) and Step 3 (glyph features), and this combined feature set was the direct input to the Random Forest model shown in Appendix D.

- **Font metadata features:** `total_pages`, `text_blocks`, `unique_fonts`, `dominant_font`, `dominant_font_ratio`, `font_variance`, `unique_font_sizes`, `dominant_font_size`, `multiple_fonts_per_page`, `rare_fonts_count`
- **Glyph and layout features:** `total_chars`, `avg_line_length`, `avg_word_length`, `avg_space_ratio`, `space_ratio_std`, `avg_font_size`, `font_size_std`, `avg_char_width`, `char_width_std`, `text_density`, `spacing_irregularity`, `font_size_variance`, `glyph_consistency_score`
- **Labels:** `filename`, `label`, `tampering_type`, `doc_type`

Appendix G: Research Summary Output

In this appendix, the final summary that was generated automatically at the end of the detection pipeline is reproduced. This summary was used as the basis for the results presented and discussed in Chapter 4.

```
PDF TEXT TAMPERING DETECTION - RESEARCH SUMMARY
Hybrid Font Metadata and Glyph Matching Algorithm
============================================================

DATASET OVERVIEW
Total PDFs: 2500
Genuine Documents: 500
Tampered Documents: 2000

TAMPERING METHODS
font_change: 500 PDFs
character_replace: 500 PDFs
spacing_alter: 500 PDFs
embedding_mismatch: 500 PDFs
```

```
DETECTION RESULTS
Rule-Based Detection Accuracy: 24.4%
Machine Learning Accuracy: 82.8%

PERFORMANCE BY DOCUMENT TYPE
invoice: 80.0%
bank_statement: 80.0%
vehicle_registration: 80.0%
legal_contract: 60.0%
academic_certificate: 80.0%

PERFORMANCE BY TAMPERING TYPE
font_change: 100.0%
character_replace: 78.0%
spacing_alter: 78.0%
embedding_mismatch: 100.0%

KEY FINDINGS
1. Font consistency is the strongest indicator of tampering
2. Embedding mismatch tampering is easiest to detect
3. Character replacement requires more sophisticated detection
4. Machine learning improves detection accuracy
```